

Use Case Flow

Student Record System

UC1 — Record Attendance (FR-001)

Goal	Record attendance for students for a selected subject.
Actor	Faculty
<i>Main flow</i>	
Steps	1. Faculty selects to record attendance. 2. System prompts faculty to select the subject and date. 3. Faculty selects the subject and date. 4. System displays the enrolled students. 5. Faculty marks each student as Present or Absent. 6. Faculty submits the attendance. 7. System validates the attendance details. 8. System saves the attendance record. 9. System confirms that attendance has been recorded.
<i>Alternate flow</i>	
Attendance not completely marked	5a. System detects that attendance status is missing for one or more students. 5b. System displays an error and asks faculty to complete the attendance. 5c. Return to step 5.
<i>Exception flow</i>	
Save failure	8a. Attendance record fails to save. 8b. System displays an error and asks faculty to retry. 8c. Return to step 6.

UC2 — Calculate Attendance Percentage (FR-002)

Goal	Calculate the attendance percentage of a student.
Actor	System
<i>Main flow</i>	
Steps	1. System retrieves the total number of classes conducted. 2. System retrieves the number of classes attended by the student. 3. System validates the retrieved attendance data. 4. System calculates the attendance percentage using the attendance data. 5. System stores the calculated attendance percentage. 6. System displays the calculated attendance percentage.

<i>Alternate flow</i>	
No attendance data available	3a. System displays a message that attendance data is not available. 3b. Return to step 1.
<i>Exception flow</i>	
Calculation failure	4a. System fails to calculate the attendance percentage. 4b. System displays an error message. 4c. Return to step 3.

UC3 — Enter / Update Marks (FR-003)

Goal	Enter or update marks for a student.
Actor	Faculty
<i>Main flow</i>	
Steps	1. Faculty selects to enter or update marks. 2. System prompts faculty to select the subject and student. 3. Faculty selects the subject and student. 4. System retrieves the existing marks, if available. 5. Faculty enters or updates the marks. 6. System validates the entered marks. 7. System saves the marks. 8. System confirms that the marks have been saved.
<i>Alternate flow</i>	
Invalid marks	6a. System displays an error and asks faculty to re-enter the marks. 6b. Return to step 5.
<i>Exception flow</i>	
Save failure	7a. Marks fail to save. 7b. System displays an error. 7c. Return to step 7.

UC4 — Calculate GPA (FR-004)

Goal	Calculate the GPA of a student based on marks and subject credits.
Actor	System
<i>Main flow</i>	
Steps	1. System retrieves the marks obtained by the student. 2. System retrieves the corresponding credits for each subject.

	3. System validates the marks and credits. 4. System converts the marks into the corresponding grade points. 5. System calculates the GPA using the grade points and credits. 6. System stores the calculated GPA. 7. System displays the calculated GPA.
<i>Alternate flow</i>	
Missing marks or credits	3a. System detects that marks or credits are missing for a subject. 3b. System displays a message indicating that the required information is unavailable. 3c. Return to step 1.
<i>Exception flow</i>	
Invalid marks or credits	3a. System detects invalid marks or credits. 3b. System displays an error message. 3c. Return to step 3.

UC5 — View My Records (FR-005)

Goal	Allow a student to view their academic records.
Actor	Student
<i>Main flow</i>	
Steps	1. Student selects to view academic records. 2. System requests the type of record to be viewed. 3. Student selects the required record. 4. System retrieves the student's academic information. 5. System displays the selected record, including attendance, marks, grades, GPA, and CGPA.
<i>Alternate flow</i>	
No records available	4a. System displays a message that no records are available. 4b. Return to step 3.
<i>Exception flow</i>	
Unauthorized access	4a. System detects that the student is not authorized to access the requested record. 4b. System denies access and displays an error message.